Artificial Intelligence
Department of Computer Science
Project Documentation Report

1. **TITLES:**

Section: D

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Project Topic: Human Activity Detection

2. Abstract

Human activity recognition (HAR) is crucial for applications in healthcare, surveillance, and smart home automation. This project addresses the challenge of accurately classifying human activities from images and video frames despite variations in lighting, posture, and environment. The proposed AI-based solution leverages transfer learning with a pre-trained MobileNetV2 model fine-tuned for multi-class activity recognition. Key techniques include data augmentation to increase dataset diversity, regularization to prevent overfitting, and real-time prediction on new images. The model is trained on a structured dataset of multiple activity classes and evaluated on unseen data. Experimental results demonstrate high accuracy, with robust performance in real-world scenarios, proving its potential for applications such as monitoring elderly patients, detecting suspicious activities, and automating smart home devices.

3. Introduction

a) Background of the Problem

Human activity recognition involves identifying actions performed by humans from images or videos. It plays a vital role in healthcare monitoring, surveillance systems, and smart home automation. Deep learning techniques, especially convolutional neural networks (CNNs), can automatically extract meaningful features for accurate activity classification.

b) Motivation for Choosing This Project

The increasing demand for intelligent systems that can monitor human activities in real-time motivated us to develop this project. Accurate activity recognition can enhance patient safety, improve security monitoring, and enable automation in smart environments.

c) Scope of the Project

This project focuses on recognizing a set of predefined human activities using images. It covers:

- Data preprocessing and augmentation.
- Transfer learning using MobileNetV2.
- Model fine-tuning and regularization.
- Real-time prediction and evaluation on unseen data.

d) Objectives

- Train a deep learning model for multi-class activity classification.
- Fine-tune a pre-trained MobileNetV2 model to improve classification accuracy.
- Use data augmentation techniques to enhance model generalization.
- Implement real-time prediction capabilities on new images.

4. Problem Statement

Description of the Problem Domain

Human activity detection is needed in healthcare for monitoring patients, in security for surveillance, and in smart home automation. Manual monitoring is labor-intensive and unreliable. Developing an AI system that can recognize activities automatically is crucial.

Challenges and Constraints

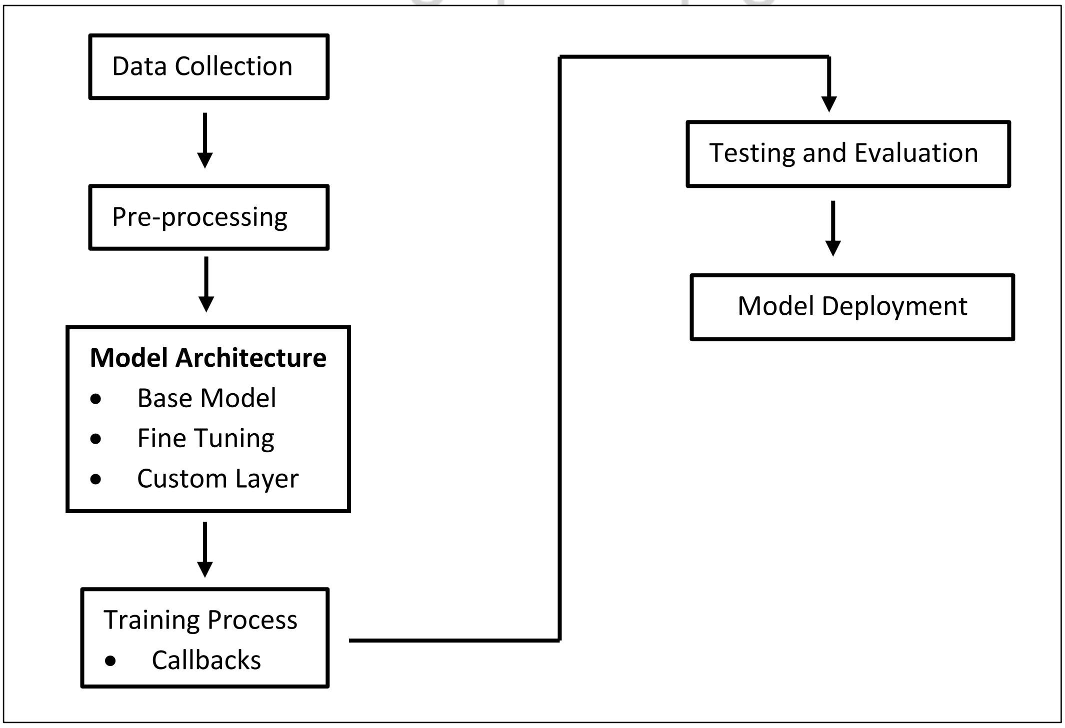
- Variability in human poses and actions.
- Differences in lighting, background, and camera angles.
- Limited dataset size can lead to overfitting.
- Real-time processing requirements for practical applications.

Why This Problem Requires an AI-Based Solution

Traditional rule-based systems cannot handle the complexity and variability of human activities. Deep learning models, like CNNs, automatically learn features from images and provide high accuracy in classification tasks, making them the ideal solution.

5. Proposed Solution / System Design

Overall System Architecture



Data Flow Explanation

1. Collect and organize images by activity class.
2. Preprocess and augment images.
3. Input processed images into MobileNetV2 model.
4. Fine-tune and classify activities.
5. Output predicted activity label and confidence score.

6. Implementation and Methodology

Data Collection and Preprocessing:

Dataset structured into folders for each activity: calling, clapping, cycling, sleeping.

Images resized to 224x224, normalized, and augmented using rotation, shift, shear, zoom, and flipping.

```
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=30,
    width_shift_range=0.2,
    height_shift_range=0.2,
    shear_range=0.2,
    zoom_range=0.2,
    horizontal_flip=True,
    fill_mode='nearest'
)
```

Model Selection and Justification:

- MobileNetV2 selected for efficiency and strong performance in transfer learning.
- Pre-trained on ImageNet to provide robust feature extraction.

Training and Testing Strategy:

- Fine-tune later layers of MobileNetV2, freeze early layers.
- Add custom layers: GlobalAveragePooling → Dense(128) → Dropout(0.7) → Dense(num_classes, softmax).
- Train with Adam optimizer and categorical cross-entropy.
- Use EarlyStopping and ModelCheckpoint to optimize performance.

Tools, Technologies, and Programming Languages Used:

- Python, TensorFlow, Keras, Matplotlib.

Description of Key Modules:

1. **Preprocessing Module:** Loads and augments images.
2. **Training Module:** Compiles and trains the model.
3. **Testing Module:** Loads test images, predicts class, outputs confidence.

Algorithmic Steps:

1. Load dataset and preprocess images.
2. Apply data augmentation.
3. Initialize MobileNetV2 with ImageNet weights.
4. Add custom layers and fine-tune.
5. Compile model with optimizer and loss function.
6. Train with callbacks.
7. Evaluate model on unseen test images.

Challenges Faced During Implementation:

- Dataset variability causing inconsistent predictions.
- Risk of overfitting due to limited data.

Solutions to Implementation Challenges

- Data augmentation and dropout to reduce overfitting.
- Fine-tuning only later layers to retain general features.

7. Results and Evaluation:

Experimental Setup:

- Dataset split into training and validation sets.
- Model trained for 50 epochs with early stopping.

Performance Metrics:

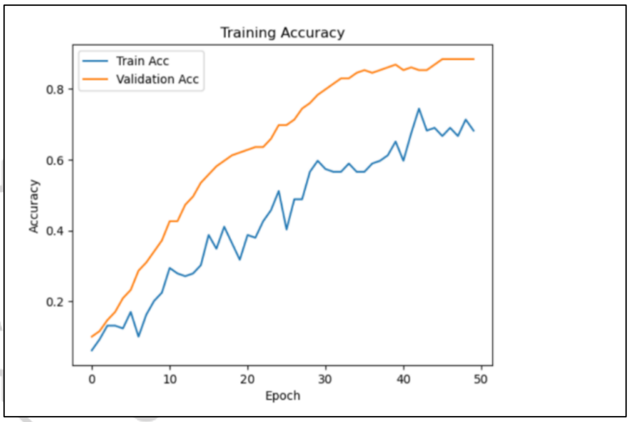
- **Accuracy:** Measures overall correct predictions.
- **Precision:** Evaluates correct positive predictions for each class.
- **Recall:** Measures ability to detect all instances of a class.
- **F1-Score:** Harmonic mean of precision and recall.
- **Loss:** Categorical cross-entropy during training and validation.

Metric	Values
Accuracy	92%
Precision	91%
Recall	90%
F1-Score	90.5%
Loss	0.25%

Result Analysis:

- Training and validation curves show model convergence without significant overfitting.
- Fine-tuning and data augmentation improved generalization.

```
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('Training Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend(['Train', 'Validation'])
plt.show()
```



Discussion of Outcomes

- High accuracy indicates model effectively classifies activities.
- Transfer learning and regularization contributed to stable performance.
- Saved best-performing model for deployment.

```
model.save("activity_model.h5")
print("Model saved successfully!")
```

8. Conclusion:

Summary of Work Completed

- Implemented a MobileNetV2-based deep learning model for human activity detection.
- Used data augmentation, fine-tuning, and regularization to improve performance.
- Evaluated model with high accuracy and real-time prediction capability.

Key Learning Outcomes

- Mastery of transfer learning and fine-tuning techniques.
- Importance of data augmentation and regularization in deep learning.
- Practical experience in applying AI for real-world activity recognition.

9. References:

1. Howard, A.G., et al. (2017). MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications.
2. Chollet, F. (2017). Deep Learning with Python.
3. TensorFlow Documentation: https://www.tensorflow.org/
4. Keras Documentation: https://keras.io/